

Artificial Intelligence

BCSE306L

Note: The content of the slides got from various sources like Internet, ChatGPT, Prescribed Textbook (Russel and Norvig), etc.

Course Objectives

- To impart artificial intelligence principles, techniques and its history.
- To assess the applicability, strengths, and weaknesses of the basic knowledge representation, problem solving, and learning methods in solving engineering problems.
- To develop intelligent systems by assembling solutions to concrete computational problems.

Course Outcomes

On completion of this course, student should be able to:

- Evaluate Artificial Intelligence (AI) methods and describe their foundations.
- Apply basic principles of AI in solutions that require problem-solving, inference, perception, knowledge representation and learning.
- Demonstrate knowledge of reasoning, uncertainty, and knowledge representation for solving real-world problems.
- Analyze and illustrate how search algorithms play a vital role in problem-solving.

Rubrics for Evaluation

1	Digital Assignment - Digital Assignment - I
2	QUIZ - Quiz - I
3	QUIZ - Quiz - II
4	CAT - Continuous Assessment Test - I
5	CAT - Continuous Assessment Test - II
6	FAT - Final Assessment Test

Textbook

- **Russell, S. and Norvig, P. 2015. Artificial Intelligence - A Modern Approach, 3rd Edition, Prentice Hall.**
- **Fundamentals of Artificial Intelligence By K.R. Chowdhary**

Intelligence

- Intelligence is the ability to learn, understand, and apply knowledge and skills. It involves the capacity to reason, plan, solve problems, think abstractly, comprehend complex ideas, learn quickly, and learn from experience.

Artificial Intelligence (AI)

- Artificial Intelligence (AI) is the simulation of human intelligence processes by machines, especially computer systems.
- These processes include learning (the acquisition of information and rules for using the information), reasoning (using rules to reach approximate or definite conclusions), and self-correction.
- AI applications include expert systems, natural language processing, speech recognition, and machine vision, etc.
- Some of the existing examples includes...

- **Expert Systems for Healthcare Diagnostic s**
- Example: **IBM Watson for Oncology:** It assists healthcare professionals in **diagnosing and treating cancer**. The AI system analyzes patient data, medical literature, and treatment guidelines to provide evidence-based treatment recommendations.
- **Natural Language Processing (NLP)**
- Example: **OpenAI's GPT-4Description:** GPT-4 is **a language model** developed by OpenAI. It can understand and generate human-like text, enabling applications like chatbots, translation services, and content creation.

- **Speech Recognition**

- Example: **Google Assistant Description:**

- Google Assistant uses speech recognition to understand and respond to voice commands. It can perform tasks such as setting reminders, playing music, and providing information based on user queries.

- **Machine Vision**

- Example: **Tesla's Autopilot Description:** Tesla's Autopilot system uses machine vision to interpret and respond to visual information from the car's surroundings. It helps in driving tasks such as lane keeping, parking, and recognizing traffic signals and obstacles.

Alexa (Amazon Alexa)

- **Natural Language Processing (NLP)**
 - **Description:** Alexa uses NLP to understand and process user queries spoken in natural language. It can perform a wide range of tasks such as playing music, setting reminders, providing weather updates, and controlling smart home devices.
- **Speech Recognition**
 - **Description:** Alexa's speech recognition capability allows it to understand voice commands from users. It can distinguish between different voices, enabling personalized responses and actions.

Siri (Apple Siri)

- **Natural Language Processing (NLP)**
 - **Description:** Siri utilizes NLP to interpret and respond to user commands and questions. It can manage tasks like sending messages, setting alarms, providing directions, and answering general knowledge questions.
- **Speech Recognition**
 - **Description:** Siri's speech recognition technology enables it to accurately understand and transcribe spoken words. This allows users to interact with their devices hands-free, using voice commands to perform various functions.

Recommendation Systems

- **Example: Netflix**
 - **Description:** Netflix uses AI-powered recommendation systems to suggest movies and TV shows to users based on their viewing history and preferences. The system analyzes user behavior and preferences to make personalized recommendations.

Autonomous Vehicles

- **Example: Waymo**
 - **Description:** Waymo, a subsidiary of Alphabet Inc., develops self-driving cars that use AI to navigate and make driving decisions. The system integrates machine vision, sensor data, and deep learning algorithms to operate vehicles autonomously.

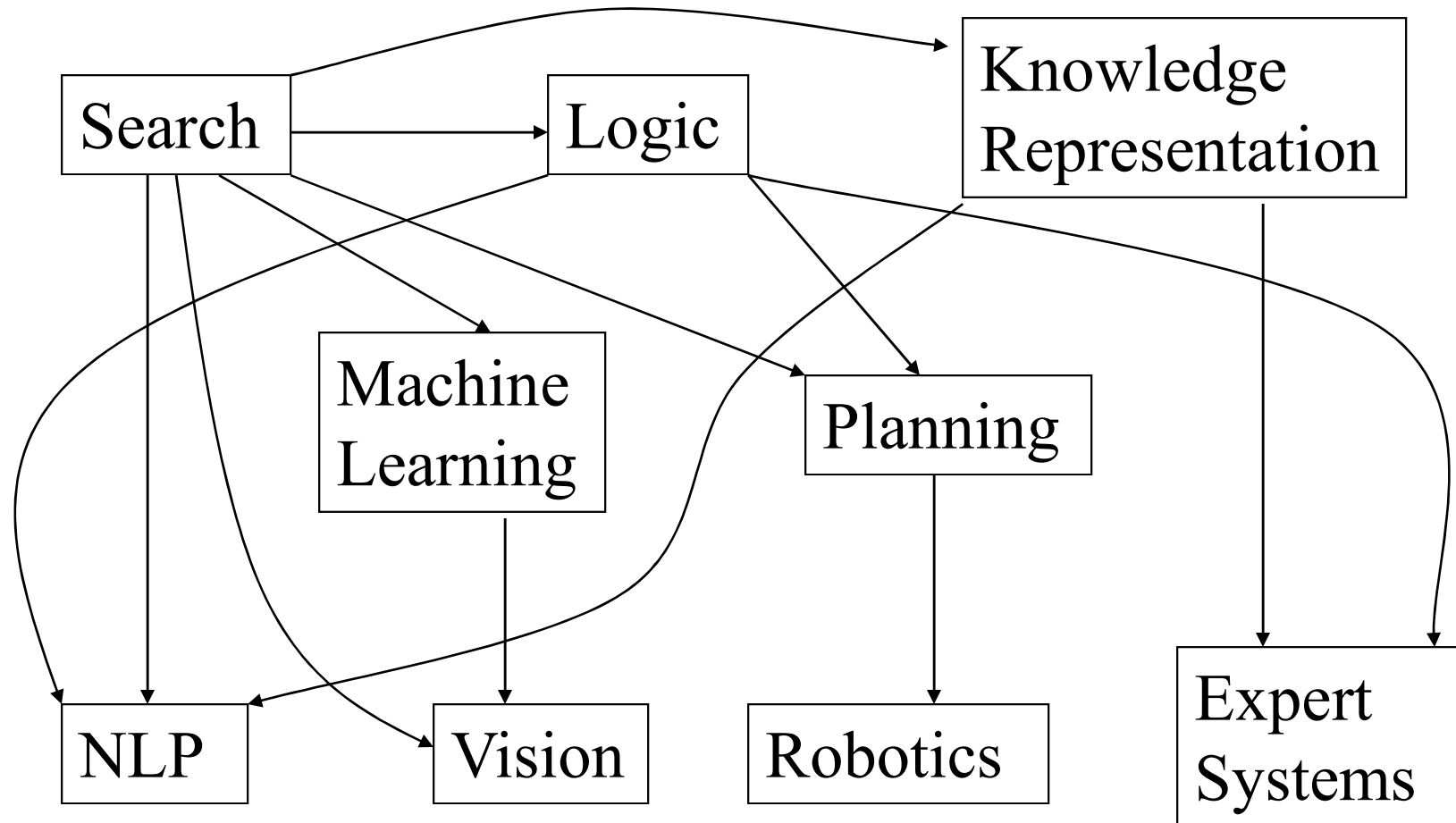
Fraud Detection

- **Example: PayPal**
 - **Description:** PayPal employs AI to detect fraudulent transactions. The AI system analyzes patterns in transaction data to identify unusual or suspicious activities and prevent potential fraud.

Image and Video Analysis

- **Example: Adobe Sensei**
 - **Description:** Adobe Sensei is an AI and machine learning platform that enhances Adobe's Creative Cloud, Document Cloud, and Experience Cloud. It can perform tasks such as auto-tagging images, improving photo quality, and generating creative content suggestions.

Areas of AI and their inter-dependencies



1. Machine Learning (ML)

- **Description:** Machine learning involves algorithms that enable computers to learn from data and make predictions or decisions without being explicitly programmed.
- **Dependencies:**
 - Relies on data collection and preprocessing techniques.
 - Often uses statistical methods and algorithms from mathematics.
 - Supports other areas like NLP, computer vision, and robotics.

2. Natural Language Processing (NLP)

- **Description:** NLP focuses on the interaction between computers and humans through natural language, enabling machines to understand, interpret, and generate human language.
- **Dependencies:**
 - Utilizes machine learning models to process and analyze text data.
 - Depends on linguistics for understanding language structure and grammar.
 - Interacts with speech recognition and generation technologies.

3. Computer Vision

- **Description:** Computer vision enables machines to interpret and make decisions based on visual data from the world, such as images and videos.
- **Dependencies:**
 - Heavily relies on deep learning, a subset of machine learning, for tasks like image recognition and object detection.
 - Uses algorithms from mathematics and statistics for image processing.
 - Can be integrated with robotics for navigation and interaction with physical environments. ↓

4. Robotics

- **Description:** Robotics involves designing, constructing, and operating robots that can perform tasks autonomously or semi-autonomously.
- **Dependencies:**
 - Uses AI for decision-making, planning, and learning from the environment.
 - Integrates computer vision for navigation and object manipulation.
 - Employs NLP for human-robot interaction.
 - Depends on hardware engineering for building physical robots.

5. Expert Systems

- **Description:** Expert systems are AI programs that emulate the decision-making ability of a human expert, often used in specific domains like medical diagnosis or financial forecasting.
- **Dependencies:**
 - Relies on a knowledge base of domain-specific information.
 - Utilizes inference engines that apply rules to derive conclusions.
 - May use NLP for user interaction and machine learning to update the knowledge base.



6. Speech Recognition and Generation

- **Description:** This area focuses on enabling machines to understand spoken language and generate speech, allowing for voice-controlled applications and systems.
- **Dependencies:**
 - Relies on NLP for understanding and processing spoken language.
 - Uses machine learning models to improve accuracy and performance.
 - Can be integrated with other areas like robotics and virtual assistants for interactive applications.

Inter-Dependencies

Data: Almost all AI areas rely on large datasets for training and improving models.

Algorithms: Shared algorithms and techniques, especially from machine learning, are foundational to various AI areas.

Integration: Real-world applications often integrate multiple AI areas. For instance, an autonomous vehicle uses computer vision for perception, NLP for voice commands, and robotics for physical control.

Advancements: Progress in one area can drive advancements in others. Improvements in deep learning techniques have significantly impacted NLP, computer vision, and more.

- These inter-dependencies highlight the collaborative nature of AI, where advancements in one area can enhance and enable innovations in others.

View's of AI

Systems that think like humans

For example, how to automate activities traditionally associated with human thinking such as decision-making, problem solving, learning (Bellman 1978)

Systems that think rationally

For example, how to use computational models to make it possible to perceive, reason and act (Winston 1992)

Systems that act like humans

For example, how to make computers do things which, at the moment, people are better at (Rich and Knight 1991)

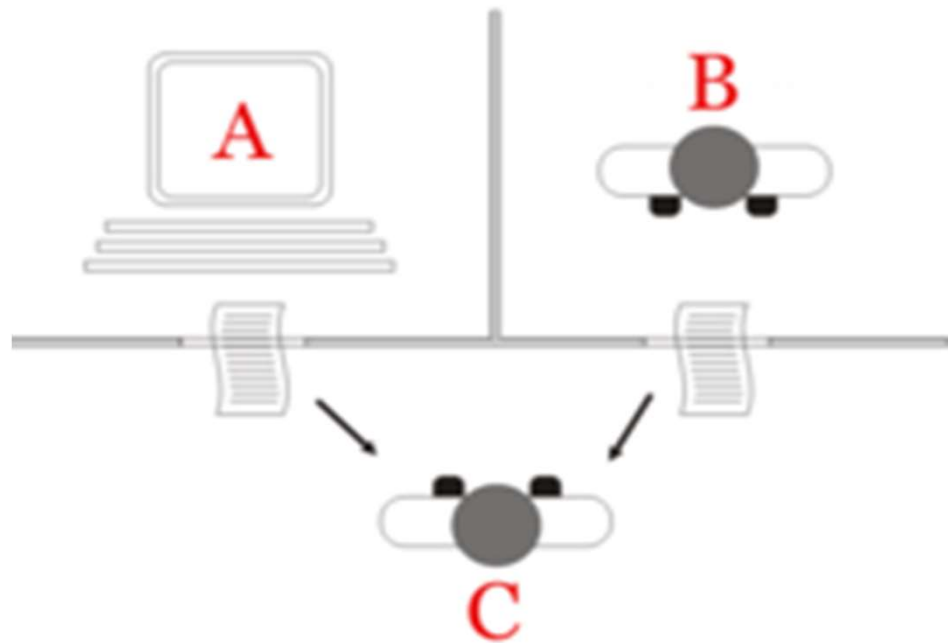
Systems that act rationally

For example, how to design intelligent agents (Poole et al 1998)

- **Systems that Think Like Humans :** AI systems that attempt to mimic human cognitive processes.
- **Systems that Act Like Humans:** AI systems that exhibit behavior similar to human actions.
- **Systems that Think Rationally:** AI systems that use logic to reason and solve problems.
- **Systems that Act Rationally:** AI systems that behave in a way that is expected to achieve the best outcome.

Turing test

- The Turing test, originally called the **imitation game** by Alan Turing in 1950, is a test of a machine's ability to exhibit intelligent behaviour equivalent to, or indistinguishable from, that of a human.



AI Tasks

- AI tasks can be categorized into three main groups: mundane tasks, formal tasks, and expert tasks.
- These following examples illustrate the wide range of tasks that AI can tackle, from everyday activities and formal problem-solving to specialized expert applications.

Mundane (Everyday) Tasks

- These tasks involve activities that humans perform naturally and effortlessly but are complex for computers.
- **Natural Language Processing (NLP)**
 - **Description:** Understanding and generating human language.
 - **Examples:**
 - Chatbots and virtual assistants (e.g., Siri, Alexa).
 - Language translation (e.g., Google Translate).
 - Sentiment analysis (e.g., analyzing social media posts).

- **Computer Vision**
- **Description:** Interpreting and understanding visual information.
- **Examples:**
 - Object recognition in images (e.g., detecting faces in photos).
 - Autonomous vehicles (e.g., Tesla's self-driving cars).
 - Medical imaging analysis (e.g., detecting tumors in X-rays).

- **Robotics**
- **Description:** Enabling machines to perform physical tasks.
- **Examples:**
 - Industrial robots in manufacturing (e.g., assembly line robots).
 - Service robots (e.g., robotic vacuum cleaners like Roomba).
 - Drones for delivery or surveillance.

- **Speech Recognition and Generation**
- **Description:** Converting spoken language into text and vice versa.
- **Examples:**
 - Voice-activated assistants (e.g., Google Assistant).
 - Automated transcription services (e.g., Otter.ai).
 - Text-to-speech applications (e.g., screen readers for the visually impaired).

- **Planning and Navigation**
- **Description:** Developing AI that can plan actions and navigate environments.
- **Examples:**
 - GPS navigation systems (e.g., Google Maps).
 - Autonomous robots navigating through environments.
 - Game AI planning moves and strategies.

- **Formal Tasks**

- These tasks involve problems that require formal methods, often associated with traditional academic disciplines.

- **Theorem Proving**

- **Description:** Using AI to prove mathematical theorems.

- **Examples:**

- Automated theorem provers like Prover9.
 - Formal verification of software and hardware systems.

- **Game Playing**

- **Description:** Developing AI that can play games at a high level.
- **Examples:**
 - Chess AI (e.g., Deep Blue).
 - Go AI (e.g., AlphaGo).
 - Video game AI (e.g., NPC behavior in games).

- **Optimization**
- **Description:** Solving problems where the goal is to find the best solution from all possible solutions.
- **Examples:**
 - Route planning for logistics (e.g., UPS delivery route optimization).
 - Resource allocation in operations research.
 - Portfolio optimization in finance.

- **Automated Reasoning**
- **Description:** Using logical inference to derive conclusions from given premises.
- **Examples:**
 - Logic programming (e.g., Prolog).
 - Rule-based expert systems.

- **Expert Tasks**

- These tasks require specialized knowledge and expertise, often in specific domains.

- **Medical Diagnosis**

- **Description:** Using AI to assist in diagnosing diseases.

- **Examples:**

- IBM Watson for Oncology.
- AI systems for interpreting medical images (e.g., detecting anomalies in MRI scans).

- **Financial Analysis**
- **Description:** Using AI for analyzing financial data and making predictions.
- **Examples:**
 - Algorithmic trading systems.
 - Credit scoring models.
 - Fraud detection systems (e.g., monitoring transactions for suspicious activity).

- **Legal Reasoning**
- **Description:** Applying AI to assist in legal decision-making and document analysis.
- **Examples:**
 - Legal research tools (e.g., ROSS Intelligence).
 - Contract analysis software (e.g., Kira Systems).

- **Scientific Discovery**
- **Description:** Using AI to assist in scientific research and discovery.
- **Examples:**
 - Drug discovery (e.g., AI systems identifying potential new drugs).
 - Climate modeling and prediction.

- **Engineering Design**
- **Description:** Assisting engineers in designing complex systems and structures.
- **Examples:**
 - CAD (Computer-Aided Design) tools with AI features.
 - AI systems for optimizing engineering designs.

Application of AI for Educational Industry

- **Mundane Tasks:** Many AI applications in education fall under mundane tasks as they involve activities that humans do naturally but are complex for machines. Examples include:
 - **Natural Language Processing (NLP):** AI chatbots and virtual assistants providing student support.
 - **Speech Recognition:** Systems converting lectures into text for notes.
 - **Computer Vision:** Analyzing student engagement through video monitoring.
- **Expert Tasks:** Some applications require specialized knowledge and analysis, such as:
 - **Learning Analytics:** Using AI to identify at-risk students and suggest interventions based on data analysis.
 - **Intelligent Tutoring Systems:** Providing personalized tutoring based on cognitive models.
- **Formal Tasks:** Few applications in education are purely formal tasks, but some elements, like automated grading, involve formal methods:
 - **Automated Grading:** Using machine learning algorithms to grade essays and assignments.

Homework

- Find the History/Evolution of AI
- Explore the state of the art of AI
- Explore the Applications of AI in each Industry like Financial, Telecom, Retail, Automobile, Healthcare, Educational, etc.